Ancestral skies Grounds keeping

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General formulation

The variables, definitions and transition function used for all three models. The specififc values of $\mathbb{P}(s|C_{st})$ will be adressed for each model along with the value function used.

Sets

S	Sites
P	Species

Data

L_s	The species found at site $s \in S$
N_s	The neighbouring sites of site $s \in S$
$\mathbb{P}(s\|C_{st})$	Probability of site s perishing by $t+1$ given the state C_{st}
$\mathbb{P}(s'\|s, A_s)$	The transition function given as GenerateOutcomes giving all possible states when all possible actions are applied and the probability of them occuring.

Stages

T	Months

States

C_{st}	Condition of site $s \in S$ at the end of month $t \in T$
x_t	The collection of C_{st} $\forall s \in S$

Actions

A_s	Save site $s \in S$, limited to one per month, by setting $C_{st} = 1$

Models

For notation purposes, define $SpeciesSaved$ as folows:

$$\sum_{p \in P} \begin{cases} 1 & \text{if } \exists s \in S \mid C_{st} = 1 \text{ and } p \in L_s \\ 0 & \text{otherwise} \end{cases}$$

Com13

Begin with initial state of all site are fragile.
$\mathbb{P}(s|C_{st}) = 0.2 \ \forall s \in S$

$$V(x_t) = \begin{cases} SpeciesSaved & \text{if no fragile sites remain} \\ \max \sum \mathbb{P}(x_{t+1}) V(x_{t+1}), x_{t+1} = x_t + A_s & \text{otherwise} \end{cases}$$

Com14

Begin with initial state of all site are fragile.
The new probability of a site perishing is dependent on the neighbouring sites state.
$\mathbb{P}(s|C_{st}) = 0.2 + 0.05 * \sum_{N_s} C_{st} = -1 \; \forall s \in S$

$$V(C_{st}) = \begin{cases} SpeciesSaved & \text{if no fragile sites remain} \\ \max \sum \mathbb{P}(x_{t+1})V(x_{t+1}), x_{t+1} = x_t + A_s & \text{otherwise} \end{cases}$$

Com15

$\mathbb{P}(s|C_{st}) = 0.2 + 0.05 * \sum_{N_s} C_{st} = -1 \; \forall s \in S$

$$V(x_t) = \begin{cases} 1 & \text{no fragile states remain and all required species are saved} \\ 0 & \text{no fragile states remain and not all required species are saved} \\ \max \sum \mathbb{P}(x_{t+1})V(x_{t+1}), x_{t+1} = x_t + A_s & \text{otherwise} \end{cases}$$

A note on the final commmunication: The first attempt of the model produced a schedual that did not include all sites. To adjust it the probability was skewed to favour higher sites. The probability value did not change the result up to 7 significant values.

0.1 AI Decleration

AI was used in communication 15 to adjust the model to favour full scheduals. See below
Q. This code skips one of 9 sites in the solution to find the optimal probability of protecting the species given in survives(s). I need it to favour producing a schedual with all 9 sites. How do you suggest i do this. Do not show me the code solution, just need the idea. ${}_Pbs =$
$defcom15(s):$
$ifsin_Pbs:$
$return_Pbs[s]$
$fragile = [iforiinSitesifs[i] == 0]$
$iffragile == []:$
$ifsurvives(s):$
${}_Pbs[s] = (1, [])$
$return(1, [])$
${}_Pbs[s] = (0, [])$
$return(0, [])$
$best_sched = []$
$best_prob = 0$
$forainfragile:$
$copy_s = list(s)$
$copy_s[a] = 1$
$copy_s = tuple(copy_s)$
$outcomes = GenerateOutcomes(copy_s, probabilities(copy_s))$
$temp = 0$
$sched = []$
$best_temp = -1$
$foryinoutcomes:$
$prob, sch = com15(y[1])$
$temp+ = y[0] * prob$
$ifprob > best_temp:$
$best_temp = prob$
$sched = list(sch)$
$iftemp > best_prob:$
$best_prob = temp$
$best_sched = [a] + sched$
${}_Pbs[s] = (best_prob, best_sched)$
$return(best_prob, best_sched)$

A. It produced 3 cases but I only applied the second.
"Modify the objective to: $score = probability + \epsilon \cdot schedule_length$
This is a standard trick in DP when you want to bias toward longer or shorter solutions without changing the primary objective."